Fairness-Constrained Allocation of Educational Support in Rural Chattogram: Pricing the Trade-off Between Predictive Reach and Demographic Coverage

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Repository: https://github.com/rafi-math/edu-support-optimization-bd
Live Project Site: https://rafi-math.github.io/edu-support-optimization-bd/
Reproduction Notebook: https://colab.research.google.com/github/rafi-math/edu-support-optimization-bd/blob/main/docs/phase3_reproduce_demo.ipynb

Abstract

Allocating limited educational support to children at the highest risk of dropping out is a central problem in education policy. While predictive models can identify vulnerable children, optimizing purely for predictive accuracy often produces allocations that are demographically skewed, systematically excluding younger children. This report estimates out-of-school risk for 4,861 children aged 6–17 in rural Chattogram, Bangladesh, using verified MICS 2019 microdata, and formulates the support allocation as a mixed-integer linear program (MILP). We price the trade-off between predictive reach and explicit demographic coverage constraints for age and gender. The unconstrained risk ranking captures 51.8% of observed out-of-school children but allocates only 0.14% of resources to children under 13. We find that age coverage is expensive, while gender parity is comparatively cheap. Furthermore, the two constraints are subadditive in objective cost (interaction −5.74 at the proportional corner): imposing both is cheaper than the sum of their individual costs. All aggregate results and reproduction code are published to ensure full reproducibility without exposing child-level microdata.

Keywords: Operations research, educational support, mixed-integer linear programming, algorithmic fairness, survey microdata.

1 Introduction

Targeting educational support to the children who need it most is a core challenge for policymakers in developing regions. In rural Bangladesh, out-of-school rates remain a persistent barrier to human capital development. Predictive models can identify children at the highest risk of dropping out, but optimizing purely for predictive accuracy often produces allocations that are demographically skewed. If out-of-school risk correlates strongly with age, a pure risk-ranking

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model will concentrate resources on older adolescents and systematically exclude younger children, even if those younger children face severe structural disadvantages.

This report examines the trade-off between predictive reach and explicit demographic fairness in the allocation of educational support. Using verified microdata from the Bangladesh Multiple Indicator Cluster Survey (MICS) 2019, we construct an out-of-fold risk score for children in rural Chattogram and formulate the allocation as a budget-constrained MILP. We systematically price the cost of imposing lower-bound coverage floors for children under 13 and for girls. The primary contribution is the empirical demonstration that these two fairness constraints are subadditive in objective cost: imposing both constraints simultaneously costs less in predictive reach than the sum of their individual costs.

2 Data and Preprocessing

The analysis uses the children's file (`fs.sav`) from the Bangladesh MICS 2019. The sample is restricted to rural households in the Chattogram Division with children aged 6–17 who have valid school attendance responses.

The raw rural cohort contains 4,891 children. The modeling-ready sample excludes 30 children due to incomplete interviews, zero survey weights, or missing core covariates, leaving a final sample of $N = 4,861$ children. The target variable is out-of-school status in the current school year. In the modeling-ready sample, there are 641 observed out-of-school children, yielding an unweighted out-of-school rate of 13.19% and a provisional child-weighted rate of 14.52%.

2.1 Data Audit and Corrections

Rigorous data validation is critical when working with complex survey microdata. During the preparation phase, several discrepancies in early processed exports were identified and corrected:

- **Weight Variable Mislabeling:** An early processed export labeled its weight column `FSWEIGHT`; however, verification against the raw metadata confirmed this column actually held raw `fshweight` values. The optimization input was rebuilt using the correct primary child weight (`fsweight`), with `fshweight` retained as a normalized sensitivity weight.
- **Non-response Indicators:** A column originally named `books_missing` was verified to be a non-response indicator for the variable `PR3`, rather than a true indicator of missing books. It was recoded accordingly to prevent the model from learning spurious signals.
- **Score Reproducibility:** The initial allocation score was not locally reproducible from the raw data. The score was entirely rebuilt from verified raw data before any fairness results were published.

3 Methodology

3.1 Risk Prediction Model

A logistic regression model predicts out-of-school status using complete core covariates: age, sex, wealth quintile, mother's education, ethnic-minority status, disability flag, and child work. To prevent test leakage, the risk score used for optimization is strictly out-of-fold, produced by 5-fold cross-validation grouped by survey cluster (PSU). This ensures no child is scored by a model fitted on that child's data.

The model is fitted without sample weights; survey weights are used only for evaluation. Logistic regression was selected over random forest and MLP alternatives based on out-of-fold performance. The out-of-fold ROC-AUC is 0.8398 (unweighted) and 0.8520 (provisional child-weighted).

3.2 Optimization Formulation

Let $x_i \in \{0, 1\}$ indicate whether child i is selected for support, and let r_i be the out-of-fold predicted risk score. The baseline allocation solves:

$$\max_x \sum_{i=1}^{N} r_i x_i \quad \text{subject to} \quad \sum_{i=1}^{N} x_i = K \tag{1}$$

where $N = 4,861$ is the sample size and $K = 730$ is the fixed support budget (approximately 15% of the sample).

To enforce demographic coverage, we introduce lower-bound constraints. Let $\mathcal{Y}$ be the set of children under 13, and $\mathcal{G}$ be the set of girls. The fairness-constrained problem adds:

$$\sum_{i \in \mathcal{Y}} x_i \geq y_{\min}, \quad \sum_{i \in \mathcal{G}} x_i \geq g_{\min} \tag{2}$$

The integer programs are solved to proven optimality (relative MIP gap zero) using the HiGHS solver via SciPy's MILP interface.

4 Results

4.1 Baseline Allocation

The unconstrained (raw risk) allocation captures 332 of the 641 observed out-of-school children (51.8%). However, it is severely age-skewed: only 1 child under 13 (0.14% of the selected set) is chosen, while girls make up 29.04%. The raw model effectively ignores younger children because out-of-school risk rises sharply with age in this cohort.

4.2 The Price of Fairness

Table 1 summarizes the policy scenarios. Imposing a proportional young floor ($y_{\min} = 423$, requiring 57.9% of the selected set to be under 13) reduces captured out-of-school children to 219, a loss of 113 children. The objective drop is 103.64 units.

Conversely, imposing population parity for girls ($g_{\min} = 367$, requiring 50.3% of the selected set to be female) costs only 12 captured children, with an objective drop of 12.91 units. Gender parity is comparatively cheap because the unconstrained model already selects girls at a 29% rate, and the risk distribution across genders is less stratified than across age bands.

Table 1: Policy scenario table. The objective is the sum of out-of-fold predicted risk scores. Observed out-of-school capture is an evaluation metric.

Scenario	Young floor	Girls floor	Young %	Girls %	Objective	Obj drop	Captured OOS	Interaction
Raw risk no floors	0	0	0.1%	29.0%	325.66	0.00	332	–
Girls 40% only	0	292	0.0%	40.0%	322.04	3.62	329	–
Girls parity only	0	367	0.0%	50.3%	312.75	12.91	320	–
Young 10% only	73	0	10.0%	25.1%	318.08	7.58	319	–
Young 20% only	146	0	20.0%	22.9%	304.27	21.39	308	–
Young 33% only	243	0	33.3%	24.2%	280.05	45.62	291	–
Young proportional	423	0	57.9%	23.4%	222.02	103.64	219	–
Young 33% + Girls 40%	243	292	33.3%	40.0%	276.47	49.19	278	−0.054
Young 33% + Girls parity	243	367	33.3%	50.3%	270.23	55.44	272	−3.095
Young prop. + Girls 40%	423	292	57.9%	40.0%	219.23	106.44	216	−0.828
Young prop. + Girls parity	423	367	57.9%	50.3%	214.85	110.81	213	−5.740

4.3 Subadditivity of Fairness Constraints

When both constraints are applied at their proportional/parity levels ($y_{\min} = 423, g_{\min} = 367$), the objective drop is 110.81. The sum of the individual drops is $103.64 + 12.91 = 116.55$. The interaction is $110.81 - 116.55 = -5.74$.

This negative interaction demonstrates subadditivity: the two fairness constraints are complements in cost. Imposing the age floor forces the model to select younger children, which alters the risk distribution of the remaining pool in a way that makes satisfying the gender floor cheaper than it would be from the unconstrained baseline. Figure 1 illustrates this geometrically; the incremental cost curves in panels C and D show that the gender floor is consistently cheaper when the proportional young floor is already active.

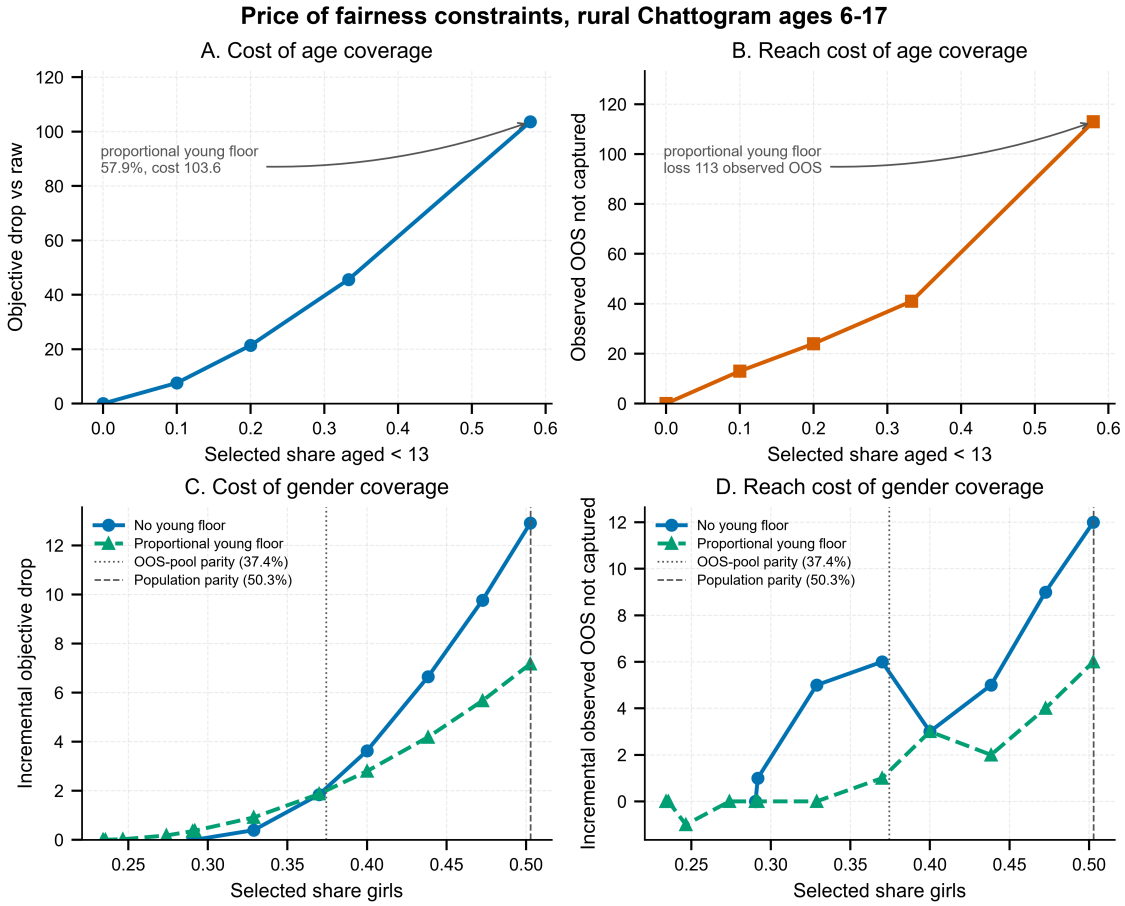


Figure 1: Price of fairness constraints. Panels A and B show the objective and reach cost of an under-13 coverage floor. Panels C and D show the incremental cost of a girls floor, with and without the proportional young floor. Grey lines indicate out-of-school pool parity (37.4%) and population parity (50.3%). The lower incremental cost of the girls floor under the young floor demonstrates subadditivity.

5 Discussion

The primary finding of this study is that demographic fairness constraints interact in non-linear ways. While age coverage is highly expensive in terms of predictive reach, gender parity is relatively cheap. More importantly, the subadditive interaction (-5.74) suggests that policymakers seeking both age and gender parity will pay a lower combined penalty than if the constraints were priced independently.

We note a critical limitation regarding spatial optimization. The MICS 2019 children's file does not contain household coordinates, school identifiers, or upazila-level travel distances. Therefore, this project does not claim to optimize physical support-center locations. Any budget or capacity value used is a scenario parameter, not an observed infrastructure variable. The optimization performed here is strictly demographic targeting.

6 Reproducibility and Privacy

MICS microdata contain household and child identifiers. Raw `.sav` files and derived child-level CSV files are kept local and are never committed to the public repository. The repository publishes only aggregate tables, aggregate curves, figures, and code.

To ensure full reproducibility without exposing microdata, the public repository includes a Google Colab notebook that clones the repository, reads the committed aggregate fairness grid, and regenerates the main four-panel figure and the policy table. The integer programs are solved with SciPy's MILP interface (HiGHS) at relative MIP gap zero, ensuring the reported allocations are proven optimal.

7 Conclusion

This report presents a reproducible framework for pricing the trade-off between predictive reach and demographic fairness in educational support allocation. By formulating the problem as a fairness-constrained MILP and evaluating it on verified survey microdata, we demonstrate that age and gender coverage constraints are subadditive. These findings provide actionable insights for policymakers designing targeted intervention programs in resource-constrained environments.

References

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